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Achieving universal electricity access in line with SDG7 using GIS-based Model: An application of OnSSET for rural electrification planning in Nigeria

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ABSTRACT

Achieving universal electricity access by 2030 is one of the energy-related development targets in Nigeria and the electricity access rate is estimated at 62%, with urban being 91% and rural 30%. In line with the set SDG7 target of achieving universal electricity access by 2030, this study examines the least-cost options for providing electricity access to thousands of unelectrified communities in Nigeria using the open source spatial electrification toolkit (OnSSET). The study focuses on the coverage area of one of the distribution companies, i.e. Kaduna Electricity Distribution Company (KEDCo) which covers Kaduna, Kebbi, Sokoto, and Zamfara States in the North-Western part of Nigeria. Spatial data covering different areas such as population, digital elevation, energy resource availability, coverage of the distribution lines, among otherswere obtained from different sources and used in the OnSSET model. The result shows that by 2030, mini-grid PV will be the least-cost technology option for 58.82% of the unelectrified communities, followed by grid-extension (20.87%), standalone PV (20.15%), and mini-grid hydro (0.17%). Further, by 2025, the total number of settlements to be electrified will be 18,182. The number of settlements electrified by 2030 will be 22,727 with an estimated population of 15.4 million. To the achieve universal electrification, a total investment of US$4.97 billion is required by 2025 with an additional US $2.88 billion by 2030. The study recommends that the state and local governments should play a key roles in providing enabling environment for community-led off-grid projects since most of the settlements require electrification projects that are less than 20 kW.

1. Introduction

1.1. Background

Access modern energy services, particularly electricity, has been recognized as a key driver of socio-economic development of countries at the micro and macro levels [1,2]. Despite the importance of electricity access, the International Energy Agency estimated that over 700 million people did not have access to electricity in 2019 [3]. The electricity access situation is worse in rural areas of Sub-Saharan Africa (SSA) whereabout 578 million people (71% of the rural population) lacked access to electricity in 2019 [3]. In recognition of the importance of energy access to the socio-economic development of developing countries, the United Nations listed access to affordable and clean energy for all as one of the 17 Sustainable Development Goals (SDGs). Goal 7 of the SDGs (SDG7) specifically states: "Ensuring access to affordable, reliable and sustainable and modern energy for all by 2030".

Achieving universal electrification, especially in rural areas, requires the combination of different technology options (grid-extension or off-grid electrification) in a manner that considers resource availability and cost [4]. The traditional method of rural electrification planning is often time-consuming and capital intensive. This is because planners will require a large set of energy-related information and data on the different factors such as socio-economic that affect the choice of adopting available technology options. Some of these data are location-specific, expensive and laborious to obtain in developing countries' settings [5–7], [8–11]. This challenge becomes even more serious in countries where several hundreds of rural communities do not have access to electricity and there is huge spatial diversity in terms of demographic attributes, energy resource availability, agro-ecological climate, geographical terrain, and social-economic conditions across such unelectrified communities.

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Advancement in geospatial technologies, especially in the area of remote sensing and geographic information system (GIS) has made it possible to overcome several data-related challenges in rural electrification planning [7]. One of the unique features of GIS-based tools is that they build on traditional energy planning methods and have the capabilities of integrating the spatial dimension of energy resources available (at a particular location), energy demand levels, existing transmission and distribution network into the rural electrification planning. The use of GIS also reduces the cost of planning, the time taken for planning and bridges some of the data-constrained situations of developing countries [12]. Based on the importance of GIS in rural electrification planning, several GIS-based tools have been developed to support rural electrification planning and applied in different national and subnational contexts. Currently, there are four major GIS-based rural electrification models that have been developed and used extensively in the literature [13]. These tools include SOLARGIS which was applied in some studies including [14–18]; Network Planner [4,19–24]; Rural electrification model (REM) [25–30]; and OpeN Source Spatial Electrification Tool (OnSSET) [7,31–42]. The review of these models can be found in Isihak et al. [13] who has carried out an in-depth review of these GIS-based rural electrification planning tools.

From the review of the four GIS-based models by Isihak et al. [13] we observe that OnSSET is increasing becoming the methodology of choice in rural electrification planning by researchers, policymakers, and rural electrification planners. This is because the tool is robust, has better capabilities and dedicated support system and community for users. The underlying grid-extension algorithms is built on an advanced technology which seems better in reflecting the on-ground reality of electrification planning. OnSSET appears to be very versatile, is open-source and accommodate more off-grid technology options. It also incorporates, as input data, more variables that influence rural electrification planning compared to other models. As such, OnSSET provide more valuable insights into decision-making for providing electricity access and takes into account more spatial nature and dynamism of the location of the settlements. In view of the versatility of OnSSET, the tool has been used extensively in the literature to provide insights and identify electrification pathways as well as addressing diverse rural electrification planning questions in countries such as Nigeria [5,13,36]; Kenya [32]; Tanzania [33,34]; Malawi [35]; Ethiopia [37]; Burkina Faso [38]; Benin [39]; Brazil [40]; Bolivia [41]; and Afghanistan [42]. Some of these studies have been done using data at the most disaggregated level (e.g. houses) while others have aggregated several smaller settlements to form larger settlements.

1.2. Case study of Nigeria

Nigeria has the highest number of persons without access to electricity in SSA. In 2019, only about 62% of the population of Nigeria had access to electricity [3,43]. Further breakdown of the electricity access rate in Nigeria shows that the access rates in urban and rural areas are 91% and 30% respectively [3]. Within the sub-national units (states), the electricity access rates vary significantly, from 10.9% in Taraba State being the state with the lowest access to 99.1% in Lagos State being the state with the highest access [44] [1] [45]. Akpan [4] noted that this disparity in electricity access rates across states is because of the difference in population density, the coverage of the existing grid, and the locations of electricity generating facilities. States with higher population density have higher commercial activities and in turn higher demand for electricity [46].

In the same vein, the Federal Government of Nigeria (FGN) has shown tremendous interest in ensuring clean and affordable energy is made available to its citizens. In 2005, the FGN, through the Electric Power Sector Reform Act (EPSRA), carried out a comprehensive reform in the electricity sector. Among other things, the EPSRA established the Rural Electrification Agency (REA) and saddled it (the REA) with the responsibility of providing affordable, reliable, sustainable, and modern energy to the rural communities through a combination of innovative renewable energy technologies in an efficient and sustainable manner [47]. Furthermore, the FGN had set an ambitious target to increase electricity access. This target is contained in the Rural Electrification Strategy and Implementation Plan (RESIP) approved by the FGN in 2016 [48]. Specifically, RESIP states that “the goal of Federal Government of Nigeria is to increase access to electricity to 75% and 90% by 2020 and 2030 respectively and at least 10% of the renewable energy mix by 2025 …“. In addition, the RESIP also states that the goal of increasing energy access “… will be achieved through: a single national, sector-wide roadmap that identifies the least-cost electrification solution for every community.” [[48], p. 9].

In line with the electricity access target and strategy set by the government of Nigeria, several studies have examined the least-cost electrification options in Nigeria at national or sub-national levels [4,5,13,19,20,36]. However, there are some research gaps in these studies. For example, previous studies have used the local government areas or electoral wards as demand nodes [4,5,19,36]. By doing this, these studies aggregated several unelectrified communities into a demand center which is not a good reflection of the peculiarities of unelectrified communities. It is likely that this was done due to the challenges of obtaining data relevant to rural electrification planning at a more granular level. Second, some of the studies [4,19,20] used Network Planner tool which has limitations in terms of the number of spatial variables that can be incorporated in the planning process as well as limitations in terms of technology options that may be incorporated into the model [49]. This study intends to address these gaps. First, this research improves on previous studies by using the most granular level of settlements data. Second, this study uses OnSSET which is a more robust rural electrification planning tool that incorporates more variables that affect the choice of electrification technology options. For example, our study incorporates data on medium voltage (MV) distribution lines, stock and locations of transformers, land use, road network, etc. within the study area [50].

Specifically, the objective of this study is to carry out a comprehensive rural electrification planning that identifies the least-cost electrification solution for every unelectrified community in our study area by 2030, using 2018 as the base year and 2025 as the step year in line with the objective of RESIP. This is done using the OnSSET. Drawing insights from Korkovelos et al [35], our research objective may be broken down into specific research questions as follows:

(i) What is the least-cost technology option for electrifying unelectrified communities within our study area?
(ii) What level of investment is required for the provision of electricity access to un-electrified communities withinour study area; and
(iii) What is the additional electricity generation capacity required to achieve universal electrification in this research study area?

The novelty of this paper is that it improves on previous studies on Nigeria (national or sub-national levels) in different areas: using data at the most granular level, and incorporating more variables into the modeling process using OnSSET.

The remaining part of the study is organized as follows: Section Two will present an overview of the OnSSET Model. In Section Three, we will present and discuss our results including the policy implications and Section Four will be the Conclusion.

[1] These are the most recent data at the state-level.

2. Methodology

2.1. OnSSET model

This study adopts the OpeN Source Spatial Electrification Toolkit (OnSSET).[2] The tool is a computational optimization decision support that identifies the least-cost electrification technology option to connect communities without electricity access and the amount of investment that will be required to deliver the projects as well as the generation capacity needed to achieve access targets. The least-cost technology option is determined by considering the levelized cost of electricity generation (LCOE) of each electrification option and selecting the option that has the lowest LCOE.

In the first stage of planning with OnSSET, the QGIS software is used to process most of the raster and vector GIS data. The outputs are then uploaded to OnSSET which is written in Python programming language and runs on python integrated development environments such as PyCharm. The output of the OnSSET can then be viewed in QGIS. The summary of the OnSSET procedure is presented in Fig. 1. The OnSSET model is described in detail in the literature [35], and on the dedicated website of the developers of the tool.[3]

2.2. Study area

This study is carried out in Nigeria. The distribution segment of the electricity sector in the country is horizontally unbundled into eleven distribution companies (DisCos) which are monopolies because they have exclusive legal right to distribute electricity in their respective franchise areas. This study is focus on the coverage area of one of such DisCos – the Kaduna Electricity Distribution Company (KEDCo), which covers four states namely: Kaduna, Kebbi, Sokoto and Zamfara as illustrated in Fig. 2. The coverage area of this DisCo is selected because of the availability of data on the coverage of the distribution infrastructure and because the states within the distribution area have relatively low electricity access rates.

The least-cost technology option for electrifying each unelectrified settlement in the study area is better understood within the context of the underlying drivers of electrification such as population distribution of the area, availability of energy resources, coverage of existing distribution network, etc. We present the characteristics of the study area relevant to our study in the following sections. We note that some of the information presented is processed using OnSSET.

2.2.1. Distribution of settlements in the study area

In this research, a settlement is defined as “a geographical area or a colony of community of people and may include hamlets, villages, towns and cities”. To obtain data on the population of settlements, we used point vector data of actual locations of settlements obtained from Bill and Melinda Gates Foundation (BMGF).[4] To make this dataset more useful for our study, we buffered the point vector by 300 m to obtain a multi-polygon feature, then we carried out an “intersection” operation with World Pop data[5] to obtain the population of the intersected points. We present the summary of the population of the settlements in the study areas in Table 1. We present the map showing settlements regarded as rural and urban in Fig. 3. Based on our data, the minimum number of people in each settlement ranges between 10 and 11 person depending on the state. The maximum number of people in the settlement's ranges from 21,584 persons in Kaduna to 49,701 persons in Sokoto State.

From Table 1, it may be observed that in the base year (2018), the total population within the study area is 11.24 million as against the 16.3 million reported by the national statistics. The difference is likely attributed to the fact that our population data are from on-ground field work by the Bill and Melinda Gate Foundation (downloaded in July 2020) and are the most granular level of settlements while the data from the National Bureau of Statistics are population projection from state-wide 2005 census figures of the states. Given that our study requires data at granular level, we elected to use the data from Bill and Melinda Gates Foundation as noted in Footnote 4.

2.2.2. Coverage of existing grid

2.2.2.1. Night-time lights. Nighttime light is a satellite imagery of the earth which shows the places on the earth that are lit at night. It is a useful dataset for understanding the current electrification status of different places. For our study, we obtained a raster map of nighttime light from Earth Observation Group, Payne Institute for Public Policy.[6] The raster map for the whole of Nigeria was clipped to the study area as shown in Fig. 4. Further, we compare the nighttime light with the raster data of population distribution obtained from World Pop (shown in Fig. 5). We observe that there are great similarities between these data which shows that places with high populations are more likely to be electrified.

2.2.2.2. Existing grid and electricity distribution assets. The coverage and stock of electricity distribution assets is very important in rural electrification planning as noted previously. Electricity distribution assets here refer to 33 kV and 11 kV low voltage distribution lines, injection substations, 33/0.415 kV transformers, and 11/0.415 kV transformers. Other supporting assets such as the different categories of electricity

[2] OnSSET is being developed by the Energy Systems Analysis Group (dESA), KTH Royal Institute of Technology, Sweden. The tool benefited from funding and support from UNESCO, UKAID-DFID, WBG, ESMAP, UNDESA, UNDP, UNECA, among others and has been applied in different countries to demonstrate its usefulness.

[3] https://onsset.readthedocs.io/en/latest/index.html.

[4] The Bill and Melinda Gates Foundation data has validated, and ground level population collected during public Health campaigns. We decided to use the data because it has names and details of real settlements and will be more useful for policy decision. We had overlaid the bill gates data on the World Pop data to assess whether the population clusters were similar. Based on our observations, we elected the approach of buffering the data and intersecting with the World Pop data.

[5] “WorldPop (www.worldpop.org - School of Geography and Environmental Science, University of Southampton; Department of Geography and Geosciences, - Funded by The Bill and Melinda Gates Foundation (OPP1134076). https://dx.doi.org/10.5258/SOTON/WP00645”. The data are in Raster format. The initial data was for the whole of Nigeria. However, we used the shapefile of the boundaries of study area to clip it.

[6] https://eogdata.mines.edu/download_dnb_composites.html.

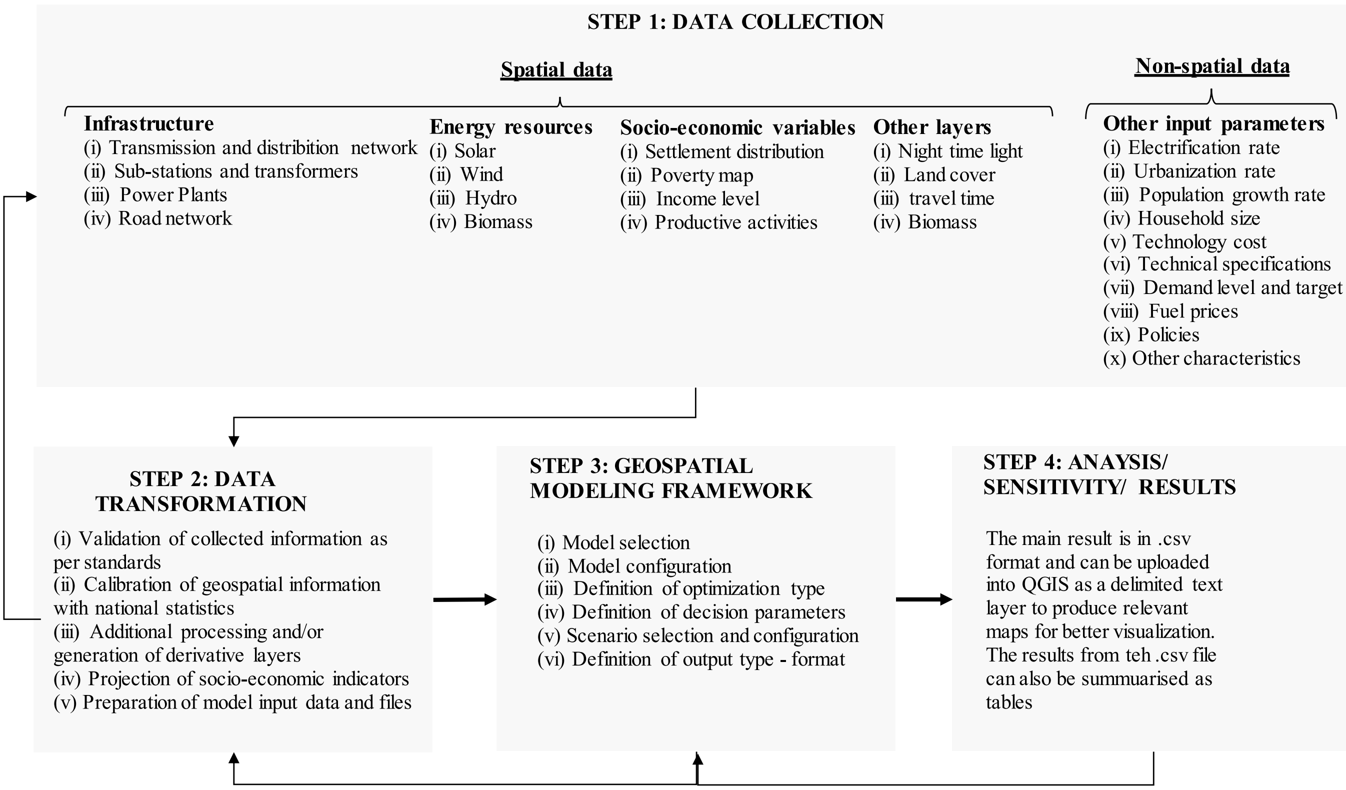

Fig. 1. Summary of OnSSET procedure. **Source:** Adapted from Ref. [35].

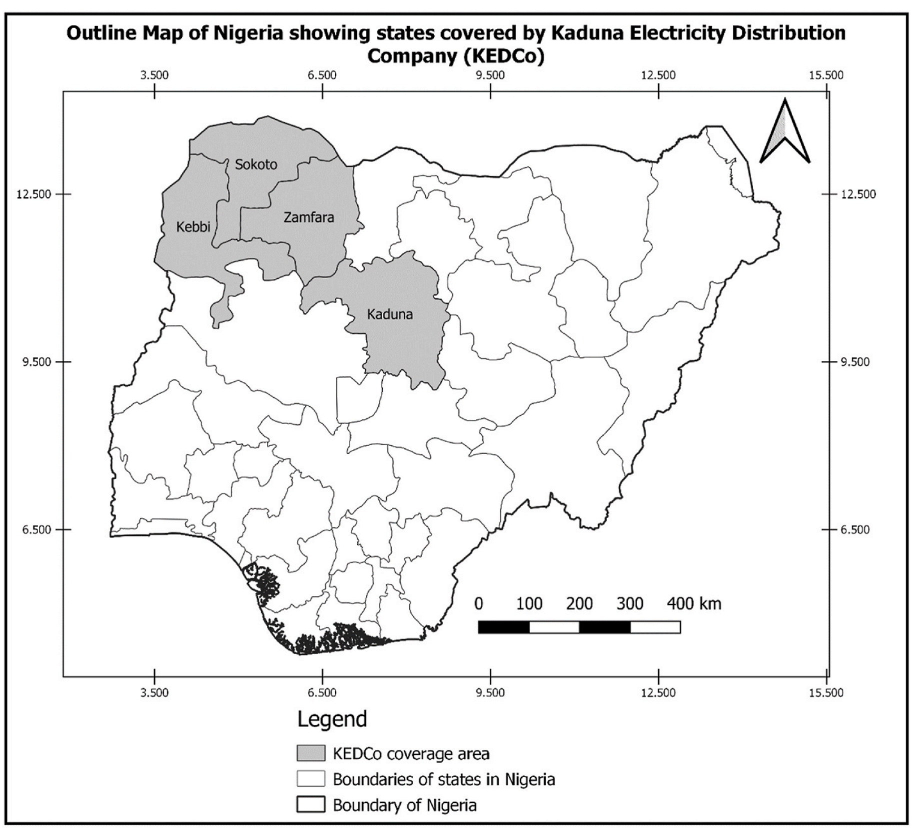

Fig. 2. Outline Map of Nigeria showing the coverage area of Kaduna Electricity Distribution Company.

Table 1
Characteristics of the settlements in the study area.

	State				Total (KEDCo)
	Kaduna	Kebbi	Sokoto	Zamfara	
Number of settlements	10,396	3,694	3,351	5,286	22,727
Rural settlements	6,432	1,597	870	2,521	11,420
Urban settlements	3,964	2,097	2,481	2,765	11,307
Min. pop of settlement	10	10	11	11	10
No. of settlements having population of 0–100	4,931	1,098	5,57	1,868	8,454
No. of settlements having population of 101–500	4,060	1,589	1,305	2,118	9,072
No. of settlements having population of 501–1000	838	461	683	658	2,640
No. of settlements having population of Above 1000	567	546	806	642	2,561
Max. pop of settlement	21,584	42,080	49,701	34,204	49,701
Ave. pop of settlement	300	628	913	518	494
Base year (i.e. 2018) population based on buffered point feature	3,116,475	2,320,755	3,058,326	2,740,088	11,235,644
2018 population (Projected from 2005 Census)	6,113,503	3,256,541	3,702,676	3,278,873	16,351,593

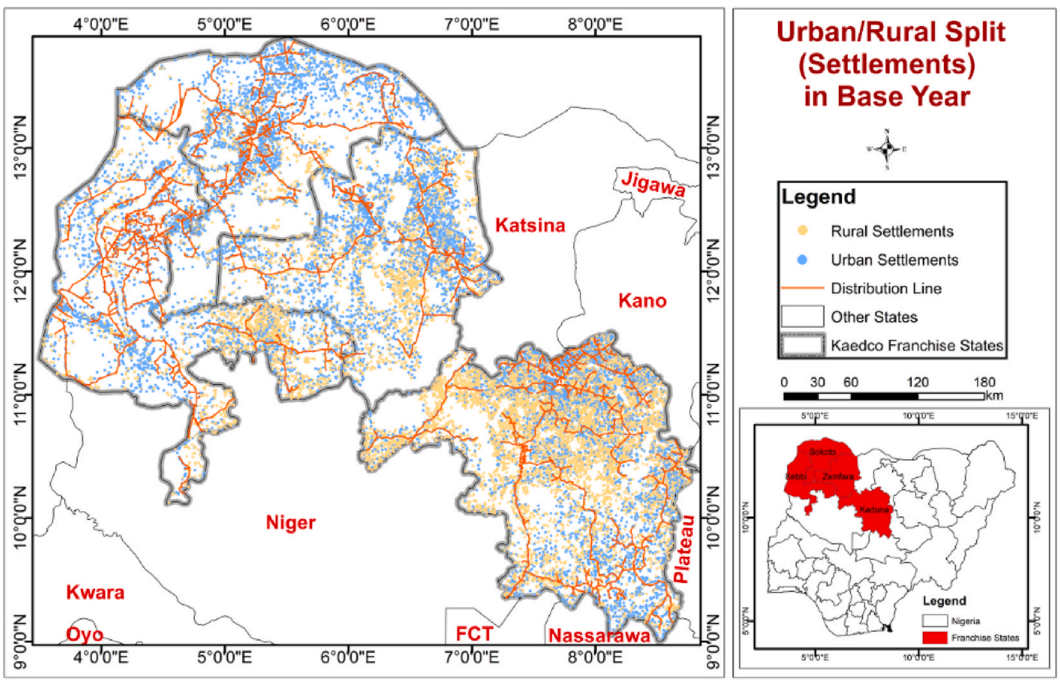

Fig. 3. Map showing the settlements that are urban and rural.

poles are also included. We obtained data on the coverage of the electricity grid in the study area from KEDCo. The dataset used in this study is the updated version of the initial data which were obtained as part of the World Bank study on Nigeria Electricity Access Project (NEAP).[7] A map showing the coverage of the distribution lines and locations of transformers is presented in Fig. 6. Comparing this map to the population distribution map in Fig. 5, we may observe that places with high populations also have a large concentration of transformers.

2.2.3. *renewable energy resources in the study area*

The availability of renewable energy resources is useful in assessing the least-cost technology option for electrifying an unelectrified community. For our study, we used data on global horizontal irradiation (kWh/m^2/year) from Global Solar Atlas[8] to evaluate the suitability of an area for solar PV systems. Data on the availability of wind resources were obtained from the Global Wind Atlas[9] while data on locations for micro-hydroelectric power were obtained from the World Bank[10]

2.2.4. *Electrification status of settlements in the study area*

Using information from the raster data of nighttime lights alongside the locations of the settlement, OnSSET identifies the settlements that are currently electrified and those that are not. These data are presented in Table 2.

The electricity access rates across the KEDCo states range between 29.1% and 53.5% based on the data from the National Demographic and Health Survey (NDHS) of 2013. We note that this is the most recent data for the rate of electricity access across the states in Nigeria. Using the OnSSET results, the ratio of the population of the electrified settlements in each state to the total population of all the settlements in the states may also be regarded as the electricity access rate. As may be observed in Table 2, this result is similar to those reported in the NDHS: 45.24% as against 53.5% for Kaduna; 46.58% as against 44.4% for Kebbi; 35.56%

[7] https://datacatalog.worldbank.org/dataset/nigeria-kaduna-electricity-distribution-company-service-area-nightlights-dataset-2016 (accessed on 28th December 2020).

[8] https://globalsolaratlas.info/download/nigeria.

[9] https://globalwindatlas.info/.

[10] https://datacatalog.worldbank.org/dataset.

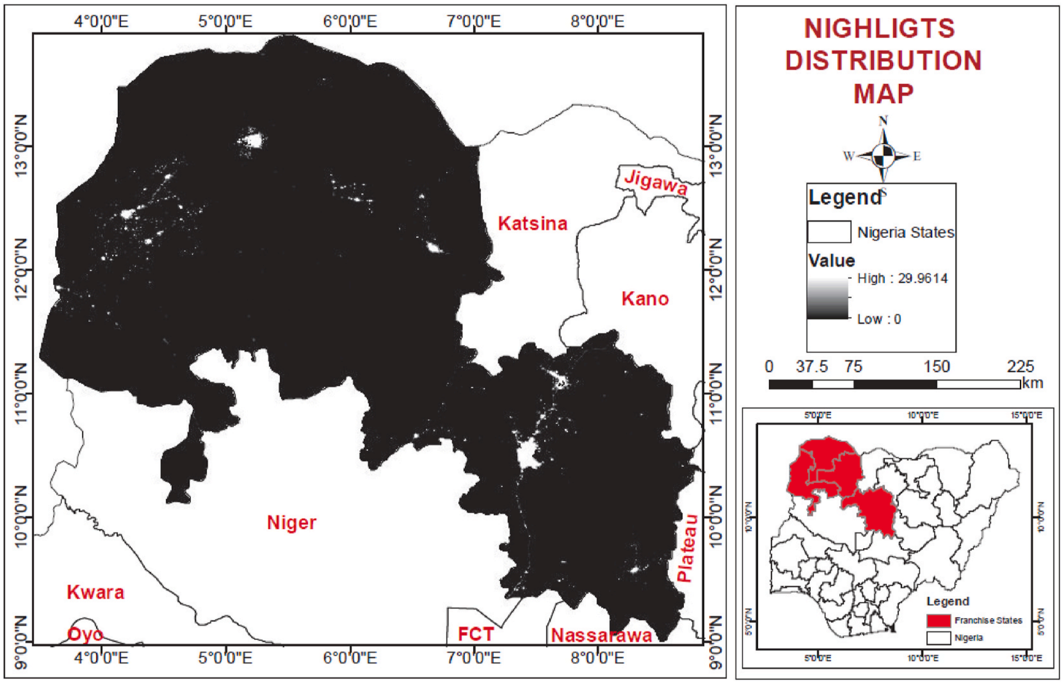

Fig. 4. Nighttime lights of the study area.

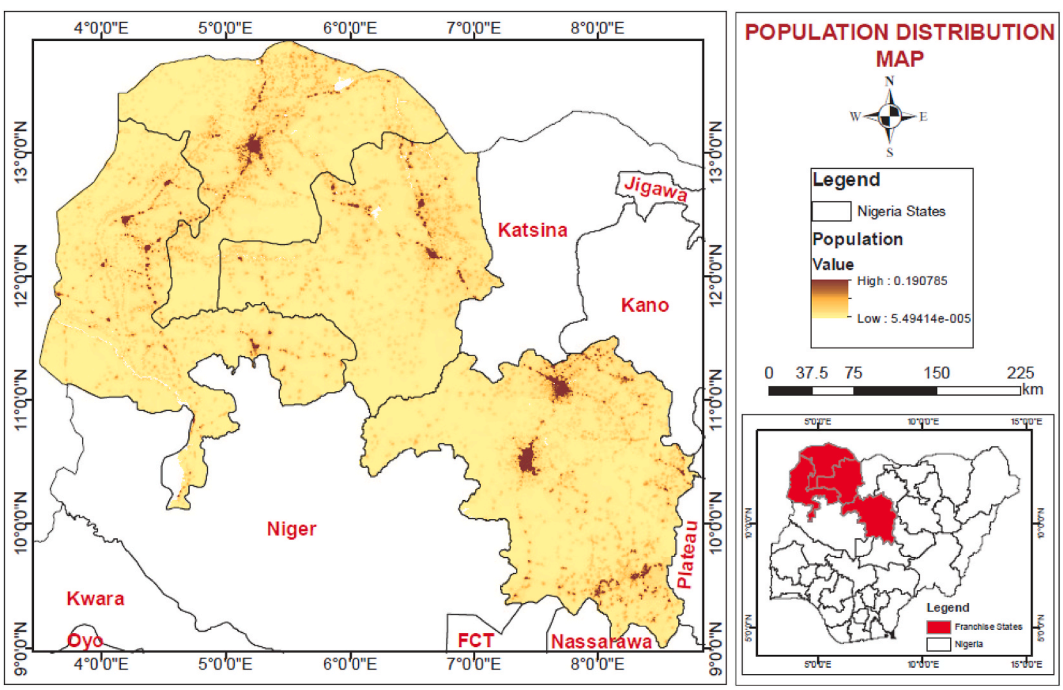

Fig. 5. Population distribution (number of persons per 1 km × 1 km grid square) of the study area based on World Pop 2020 raster data.

as against 38.9% for Sokoto; and 24.86% as against 29.1% for Zamfara.

3. Results

3.1. *Least-cost technology option for electrifying unelectrified communities in KEDCo coverage area*

Table 3 shows the result of the least-cost technology option for electrifying unelectrified communities in the study area.

We observe from Table 3 that in 2025, about 80% (i.e. 18,182 out of 22,727) of the entire settlements in the KEDCo area may be electrified while universal electrification would be achieved in 2030. We also observe that mini-grid PV is the least-cost option for electrifying 10,190 settlements (about 56.04% of the total settlements) in the 2025. The number of settlements which mini-grid PV may be the least-cost electrification option could increase from 10,190 in 2025 to 13,368 (or 58.82% of the total settlements) in 2030. The dominance of mini-grid solar as an electrification option is due to relatively high solar irradiation in the study area which reaches 6 kWh/m^2/day in some places. We further observe that in 2030, 79.14% of the settlements currently

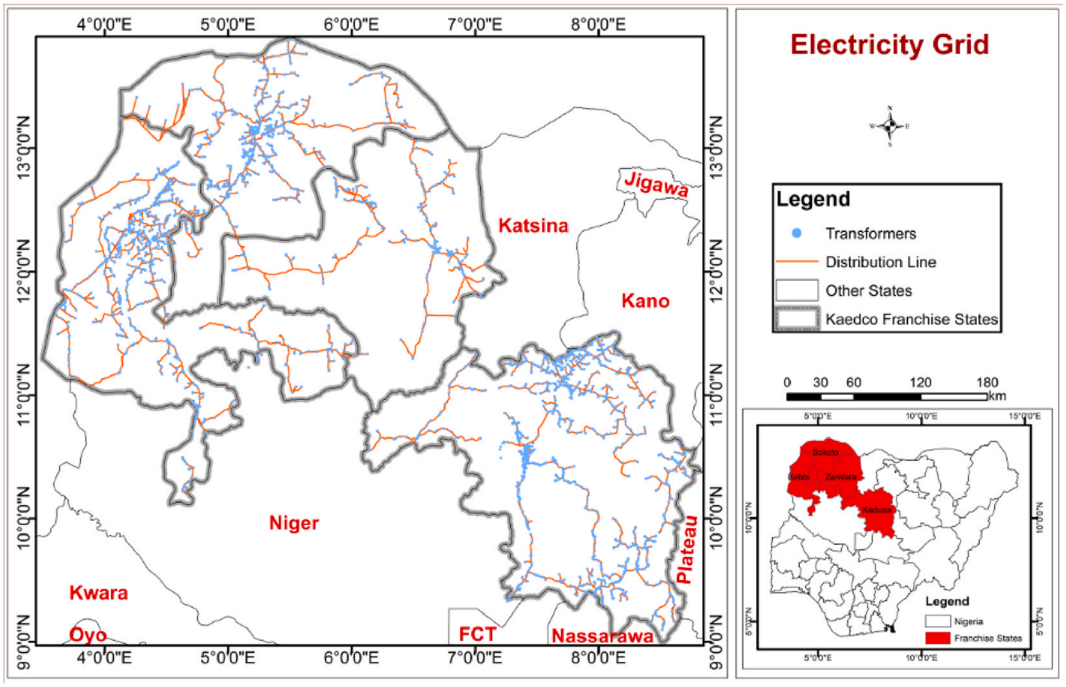

Fig. 6. Coverage of existing distribution network (11 kV and 33 kV lines, transformers) in KEDCo area. **Source of data:** Kaduna Electricity Distribution Company (KEDCo).

Table 2
2018 electrification status of the study area.

	State				Total (i.e. KEDCo)
	Kaduna	Kebbi	Sokoto	Zamfara	
No. of settlements	10,396	3,694	3,351	5,286	22,727
Population of settlements (base year)	3,116,475	2,320,755	3,058,326	2,740,088	11,235,644
No of settlements electrified (base year)	2,516	874	721	631	4,742
% of settlements electrified (base year)	24.20%	23.66%	21.52%	11.94%	20.87%
Population of the settlements electrified (base year)	1,409,752	1,081,010	1,087,687	681,138	4,259,587
% of population of settlements electrified (base year)	45.24%	46.58%	35.56%	24.86%	37.91%
Electricity access rate (as reported in NDHS)	53.5%	44.4%	38.9%	29.1%	

Table 3
Number and percentage of settlements where each technology option will be the cost-effective option in KEDCo area for the step year (i.e. 2025) and final year (i.e. 2030)

	No. of settlements electrified in 2025		No. of settlements electrified in 2030	
	Number	Percentage	Number	Percentage
Mini-grid hydro	11	0.06%	38	0.17%
Mini-grid PV	10,190	56.04%	13,368	58.82%
Standalone PV	4,187	23.03%	4,579	20.15%
Grid extension	3,794	20.87%	4,742	20.87%
TOTAL	**18,182**	100.00%	**22,727**	100.00%

without access to electricity will be compatible with off-grid renewable energy (with Mini-grid PV accounting for 58.82%, Standalone PV; 20.15% and Mini-grid hydro; 0.17%), while the remaining 20.87% of the settlements might be electrified through the grid-extension. The role of mini-grid hydro is minimal across the years which is a reflection of the location-specificity of the resource. We also present in Fig. 7 the map showing the technology options that would be cost effective for providing electricity to the populations in 2030.

One may observe from Fig. 7 that some of the mini-grid systems are close to places that are suitable for grid-extension, while others are far away. It is possible that some of the settlements that are suitable for mini-grid PV may become suitable for grid-extension if some of the parameters in the model are changed. If actual electrification projects are to be carried out in such places, it might be useful to carry out additional techno-economic evaluation to understand which option may be the least-cost one.

Furthermore, in our result, grid-extension might still play a relatively significant role in providing electricity to the unserved areas (20.87% of the communities to be electrified through grid extension). The grid-extension contribution is lower than the mini-grid because there is high cost of extending the distribution network especially given the nature of the settlement in the study area. The settlements in the KEDCo area are widely dispersed and as such, connecting with grid implies running distribution lines for long distances to serve just few population centers.

It is insightful to understand the extent to which the size of a settlement might influence which technology option is the most cost-effective one. We provide in Table 4, a breakdown of the number of settlements by population size and their compatibility with the different technology options in the end year 2030.

We understand that grid-extension is usually preferable if a settlement is in close proximity to the grid. One may infer from Table 4 that the settlements within the different population bands in the grid-extension row are relatively close to the existing grid which makes grid-extension the cost-effective option. For settlements not close to the

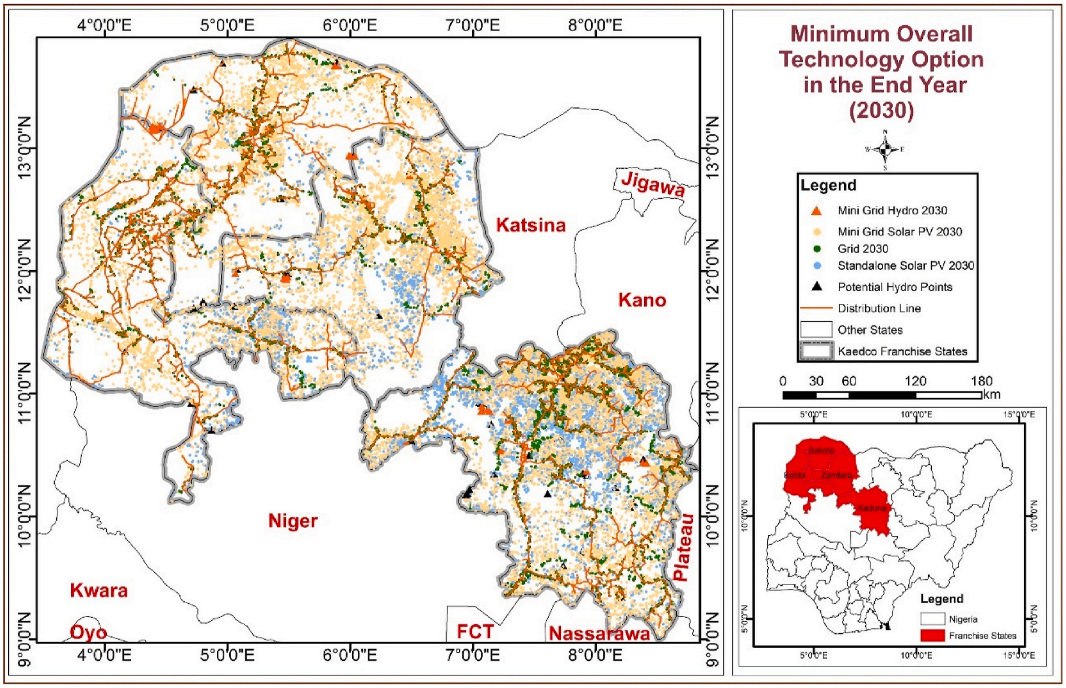

Fig. 7. Map of KEDCo area showing the cost-effective technology option for achieving universal electricity access in the final year 2030.

Table 4
Cost-effective technology options in KEDCO by sizes of settlements in 2030.

	Number of settlements (2030)						
	population: 0-50	population: 50-100	population: 100-250	population: 250-500	population: 500-1000	population: above 1000	Total
Mini-grid hydro	8	13	5	5	6	1	38
Mini-grid PV	123	2,442	4,206	2,926	2,041	1,630	13,368
Standalone PV	4,323	256	0	0	0	0	4,579
Grid extension	753	536	1,079	851	593	930	4,742
Total	**5,207**	**3,247**	**5,290**	**3,782**	**2,640**	**2,561**	**22,727**

Table 5
Summary of result for KEDCO area showing the number of settlements with the technology option that is the least–cost option as well as the population to be electrified, and investment requirements in the step year and final year.

		Mini-grid hydro	Mini-grid PV	Standalone PV	Grid extension	TOTAL
2025	No. of settlements electrified in 2025	11	10,190	4,187	3,794	18,182
	Corresponding 2025 population	1,004	7,961,879	177,961	5,314,368	13,455,212
	Investment (million US$) in 2025	0.29	4,577.40	76.90	315.36	4,969.95
	Investment per capita (US$/person)	291.28	574.91	432.11	59.34	–
2030	No. of settlements electrified in 2030	38	13,368	4,579	4,742	22,727
	Corresponding 2030 population	10,856	8,986,022	145,478	6,255,858	15,398,214
	Additional investment (million US$)	0.27	2,619.30	36.69	221.88	2,878.14
	Investment per capita (US$/person)	24.51	291.49	252.23	35.47	–

existing grid, the different off-grid technologies were considered. Since mini-grid hydro requires a location-specific energy resource, i.e. hydropower, we observe that settlements in mini-grid hydro row are closer to the required resource. For the remaining off-grid electrification options, i.e. mini-grid PV and standalone PV, we observe that standalone PV is the cost-effective option for settlements with low population while mini-grid PV becomes the cost-effective option as population increases.

3.2. Investment requirement for achieving universal electricity access in the study area in 2030

We present in Table 5 the summary of results relevant for this objective. We observe from the table that in the step year (i.e. 2025), about 13.46 million people will have access to electricity with mini-grid being the least-cost option for most of the settlements and populations (about 7.96 million people). To achieve 80% access to electricity in the study area by 2025, a total investment of US$4.97 billion will be required. In the final year 2030, an additional investment of US$2.88 billion will be required to achieve universal electrification.

By putting the investment requirement side-by-side with the population to be catered for by each technology option, it can be observed that investment per capita for grid extension is the lowest while that for mini-grid PV is the highest. This implies that if the target is to increase the number of persons that will benefit from electricity access projects, grid-extension may be the economically viable alternative.

Table 6
Summary of additional electricity generation capacity required for each technology option to achieve universal electrification in the KEDCo area in the step year (i.e. 2025) and final year (i.e. 2030)

		Mini-grid hydro	Mini-grid PV	Standalone PV	Grid extension	TOTAL
2025	No. of settlements	11	10,190	4,187	3,794	18,182
	Corresponding 2025 population	1,004	7,961,879	177,961	5,314,368	13,455,212
	New Capacity (kW)	46	233,966	1	109,548	343,561
2030	No. of settlements	38	13,368	4579	4,742	22,727
	Corresponding 2030 population	10,856	8,986,022	145,478	6,255,858	15,398,214
	New Capacity (kW)	90.74	1,378,538	14,068	89,811	1,482,510

3.3. Additional electricity generation capacity required to achieve universal electrification in the study area

The summary of results showing additional electricity generation required to achieving universal electrification in the study area is presented in Table 6.

The additional capacity required to achieve universal electricity coverage in the study area by 2030 is 1.48 GW. This is made up of 89.8 MW from grid-extension, 14.07 MW for standalone PV, 1.38 GW from mini-grid PV and 90.74 kW from mini-grid hydro. The 1.48 GW new capacity is expected to provide electricity to about 15.4 million people. Across the different technology options, we observe that the 90.74 kW new capacity for mini-grid hydro is to provide electricity access to 10,856 people; the 1.38 GW new capacity for mini-grid PV is expected to provide electricity for 8.99 million people; the 14.07 MW new capacity for standalone PV would provide electricity for 145,478 people; and the 89.81 MW new capacity for grid-extension is expected to provide electricity for 6.26 million people. It can also be observed that the total new capacity may increase fourfold from 343 MW in the step year to 1.48 GW in the final year. This increase in the new capacity requirement reflects the increase in expected electricity demand and represented in the different tiers of consumption.

Using the information in Table 6, we computed the average new capacity per settlement for each technology option as follows: mini-grid hydro – 2.35 kW; mini-grid PV – 103.12 kW; standalone PV – 3.07 kW; and grid-extension – 18.94 kW. This shows that communities that are compatible with standalone PV might require a project with capacity of 3.07 kW on average. This is within the range of standalone projects supported by the World Bank-assisted Nigeria Electrification Project (NEP). Similarly, communities that are compatible with mini-grid PV could require a mini-grid project with capacity of 103.12 kW on average.

3.3.1. Additional insights for 2030 results

We observe from Table 6 that 14,068 kW of new capacity of standalone PV might be required to provide electricity to 4,579 settlements with a population of 145,478 in 2030. This information is obtained by summing up the new capacity requirement for all the settlements compatible with standalone PV. We disaggregate the total number of settlements compatible with different technology options by population sizes to examine the roles of population in the selection of the least-cost technology options. We also disaggregate the total number of settlements to examine the relationship between technology options and capacity sizes. These results are presented in Table 7 and Table 8 respectively.

From Table 7, we observe that of the 4,579 settlements where standalone PV is the least-cost option, 4,323 have population between 0 and 50 while the remaining 256 have population between 50 and 100. Moreover, we observe from Table 8 that the new generation capacity required for the 38 settlements that were compatible with mini-grid hydro are between 0 and 20 kW. The result for standalone PV is also similar to this. For mini-grid PV, 54.6% of the settlements require new capacities that are below 20 kW. In accordance with the Mini-grid guidelines in Nigeria, any project that is generating capacity of less than 100 kW might not require permits issued by the government. The Mini-grid developer is only required to register online and proceed with project. In all, the sizes of new capacities requirement for all the technology options are less than 20 kW for 70.76% of the settlements. This implies that majority of the projects would benefits from permit exemptions.

3.4. Comparative analysis of our results with previous studies

Our result aligns with the findings of Bhattacharyya & Palit [49] who observed that models that use data at granular level and incorporate LV distributions infrastructure are likely to recommend off-grid technologies for providing electricity access to unelectrified settlements, while grid-extension tend to be recommended when data are more aggregated.

Comparing our results with those of previous studies that focused on Nigeria [4,20,51,52], we observe that our results are different from the findings of these studies. For example, the result of Ohiare [52] shows that grid-extension is the least-cost option for providing electricity access to about 98% communities in Nigeria while only 2% were to be covered by the mini-grid. The findings of Akpan [4] indicates that extending the grid is the least-cost option for 98.7% and 89.5% of the demand centers in the respective study locations; while World Bank [20] shows that grid-extension is the least-cost option for electrifying 68% communities by 2030. In contrast, our study shows that mini-grid PV is the most cost-effective option for 58.82% of the unelectrified communities in the study area. There are some inter-related factors that may be responsible for the difference in our results. First, the least-cost technology option is largely dependent on the cost of deploying the technology. The cost of solar PV systems has been declining over time which makes technologies such as mini-grid PV and standalone PV relatively more competitive. Second, the grid-extension algorithm (Kruskal's Minimum Spanning Tree algorithm) used by Network Planner is biased towards the grid-extension [4,21] as compared with OnSSET which uses a tree search algorithm complemented by Locality-Sensitive Hashing (LHS) [35].

Third, our study is carried out at a very disaggregated level: our study uses actual settlements as demand centers unlike Ohiare [52] that

Table 7
Least-cost technology options for the different settlements - disaggregated by population size – 2030.

	Pop. Size: 0-50	Pop. Size: 50-100	Pop. Size: 100-250	Pop. Size: 250-500	Pop. Size: 500-1000	Pop. Size: above 1000	Total no. of settlements
Mini-grid hydro	8	13	5	5	6	1	38
Mini-grid PV	123	2,442	4,206	2,926	2,041	1,630	13,368
Standalone PV	4,323	256	0	0	0	0	4,579
Grid extension	753	536	1,079	851	593	930	4,742
Total	**5,207**	**3,247**	**5,290**	**3,782**	**2,640**	**2,561**	**22,727**

Table 8
Least-cost technology options for the different settlements - disaggregated by generating capacities - 2030.

	No. of settlements with capacity between 0 and 20 kW	No. of settlements with capacity between 20 and 50 kW	No. of settlements with capacity between 50 and 100 kW	No. of settlements with capacity above 100 kW	Total no. of settlements (number)
Mini-grid hydro	38	0	0	0	38
Mini-grid PV	7,296	3,793	1,173	1,106	13,368
Standalone PV	4,579	0	0	0	4,579
Grid extension	4,161	0	306	275	4,742
Total	**16,074**	**3,793**	**1,479**	**1,381**	**22,727**

used local government areas which is the third administrative division in Nigeria to represent a single demand center; or Akpan [4] that used electoral wards (fourth political administrative division in Nigeria) as demand centers. It is important to note that a single local government area (LGA) in Nigeria is made up of between 10 and 20 electoral wards and a single ward may consist of tens of settlements. Some of these communities are also apart from each other by several kilometers and equally far from the national grid, therefore aggregating all the communities in a ward or local governments area into a single community will imply that the electrification model will assume a high unelectrified population and electricity demand which will tend to make grid-extension more cost-effective [53]. Fourth, OnSSET requires more spatial and energy related data which makes OnSSET analysis and result to reflect the practical realities of electrification and provide more insightful.

Another study worth comparing with our study was done by Mentis et al. [51] that carried out the first study in Nigeria where the OnSSET was applied. The results show that extending the grid is the lowest cost option for approximately 86% of the unelectrified settlements by 2030 in the whole of Nigeria. This result is also in contrast to our findings despite using the same methodology and even though it is a different case study application. The primary reason for the differences in findings is that the study used local government areas as it demand centers which will make the result to be biased towards grid-extension. Our results are similar to the results of Korkovelos et al. [35] that observed that off-grid PV was the least-cost option for 67.4% of unelectrified settlements in Malawi while grid-extension was the least-cost option in 32.6% of the total settlements. In contrast, our results differ from Balderrama et al. [41] who observed that grid-extension was the least cost option for 76.2% of electrified settlements in Bolivia; standalone PV was the least-cost option for 16.8% of such settlements; while the other technology options were the least-cost options for the remaining settlements. The results of the study by Mentis et al [37] in Ethiopia shows that grid-extension will be the best option for connecting 93.4% of unelectrified households. Similarly, the result of Korkovelos et al. [42] shows that grid-extension will be the least-cost option for electrifying 51.7% of settlements in a study area in Afghanistan while different off-grid solutions will be compatible with the remaining number of settlements.

Furthermore, we compared cost per capita of our result with previous studies. The electrification cost per capita is a unique indicator of rural electrification planning. We note that not all studies reported the electrification cost per capita. Some studies (e.g. Akpan [4]) reported the electrification cost per household as well as the average household size. The summary of electrification cost per capita as obtained (or computed) from other studies is presented in Fig. 8.

From Fig. 8, we observe that the cost per capita for grid extension for 2030 is US$35.47 for ourstudy area which is far lower than the cost per capita for grid extension estimated by Akpan [4] for Taraba and Yobe States being US$1362.64 for each state respectively. The rationale for the cost difference may include: (i) Akpan [4] was conducted when the exchange rate in Nigeria was US$1 = ₦165 whereas our study was conducted when the exchange rate was US$1 = ₦370. Therefore, since the cost of grid extension is estimated in local currency, these cost components estimated in local currency may seem to have a lower foreign currency equivalent; (ii) in our study, we use comprehensive cost of grid-extension and also provided a more robust data including the distribution lines and the stock of transformers within our study areas. As such, the OnSSET model in our analysis does not need to add these additional costs to estimate the grid extension costs unlike in all the previous studies we compared with where such key parameters are omitted, and the model appears to have added those cost elements in

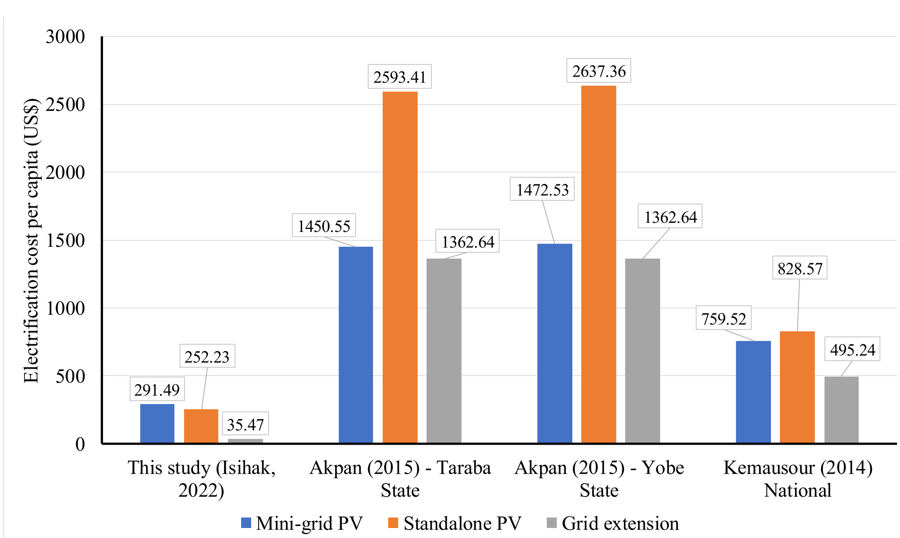

Fig. 8. Comparison of electrification cost per capita for grid-extension with those of other studies.

arriving at the final investment required to provide energy access. We also observe a similar pattern with respect the standalone PV and mini-grid PV. We note that the unit cost of solar PV system has been decreasing steadily which may contribute to the differences in cost. We also observe that the standalone PV had the highest electrification cost per capita in the studies by Akpan [4] and Kemausour [22] whereas mini-grid PV has the highest cost per capita in our study.

Furthermore, Jean et al. [40] and Bhattacharyya & Palit [49] showed that assumptions used in the OnSSET modeling process can significantly affect the results of different technology options. Changing the assumptions on some variables such as the household demand level and cost of the different technology options may affect the outcome of the study.

3.5. Discussion and policy implications

We may recall that OnSSET uses the LCOE to select the least-cost technologies. We note that the cost per kW of mini-grid PVs in developing countries is generally reducing and the cost projections to 2030 indicates that it will reduce further [54]. This may explain why mini-grid PV is the least-cost technology option for achieving universal electricity access. This also implies that mini-grid PV may even be more cost-effective in the future. The prospect of mini-grid PV in providing electricity access is in line with global trends [54]. The amount required to provide electricity access to the unelectrified communities in the study area is relatively huge. Thus, the greatest challenge in achieving the electrification target is the huge investment requirements. The fact that the entire electricity sector in Nigeria has been deregulated implies that the investment and funding that will be required will have to be private-sector driven. Currently, the electricity distribution companies in Nigeria are owned by private businesses. Based on the experience of other electricity distribution companies [4], such companies are seldom interested in rural electrification because of difficulties to realise revenue due to affordability issues among the rural people. To address the huge investment barrier as identified in this study, the Government of Nigeria, through the Rural Electrification Agency started the Nigeria Electrification Project (NEP) which is pro-poor public-private partnership (5Ps) for rural electrification. The NEP has financial support of US$ 350.00 million [55] and US$150 million [56] from the World Bank and the African Development Bank respectively. The NEP focuses on mini-grid and standalone projects which aligns with the findings of the study on the huge role of mini-grid and standalone systems in providing access to electricity.

Apart from government's partnership with multilateral development agencies for the development of mini-grids, there is also need for other forms of private sector support for delivering electricity access services in form of fiscal incentives. One of such may be a policy to ensure that commercial loans for solar mini-grid projects have longer tenor and low interest rates. There is also a need for increased awareness to ensure mini-grid and standalone projects are embarked on self-help projects by communities themselves with a clear sustainable business model. Experience has shown that self-help projects are often sustainable in the long run [57]. Moreso, continuous capacity building for government agencies overseeing the rural electrification space (i.e. the Rural Electrification Agency and the Nigeria Electricity Regulatory Commission) is also important to ensure the successful deployment of mini-grid and standalone projects in a conpetent manner.

4. Conclusion

The objective of this study was to use a GIS-based rural electrification planning tool, i.e. OnSSET, to obtain the least-cost technology option for achieving universal electrification target in 2030 in Nigeria. The study also estimated the investment requirement for achieving this target as well as the additional generation capacity that will be required. Our result shows that mini-grid PV will be the least-cost technology option for electrifying about 58% of unelectrified communities in 2030. Standalone PV and grid-extension are the least-cost technology options for 20.15% and 20.87% of the communities respectively, while mini-grid hydro plays a minor role. We also observe that some communities where mini-grid PV is the least-cost option are relatively close to the existing grid. We recommend that the type of projects to be considered in settlements where mini-grid is the least-cost option but have the potential for grid connection based on their proximity should be mini-grids where integration into the grid will require a minimal capital cost.

The investment requirement for achieving the universal electrification in the study area runs into several billions of United States dollars. The challenge is on how to generate these investments. The Rural Electrification Agency (REA) will have to play a huge role in this direction. Luckily, in recent times, the REA with support from the World Bank is experimenting with pro-poor public-private partnership (5Ps) models for rural electrification. It will be very useful to evaluate the success of these models on the various electrification projects executed by REA in conjunction with World bank to examine whether such models can be replicable or scalable to cover more communities.

Finally, the result of the study shows that renewable energy technologies (i.e. mini-grid PV, mini-grid hydro, and standalone PV) will provide electricity to about 80% of unelectrified settlements thereby playing a dominant role in achieving electricity access targets. This will also contribute and accelerate the achievement of the Nigeria's net-zero target by 2060.

Credit author statement

Salisu R. Isihak: Conceptualization, Validation, Methodology, Investigation, Writing - Original Draft,Resources,Reviewing and Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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